

Request for Information – Engineering Study on Interchangeability of Dowty R391 Propellers between C-27J and HC-130J

- Background: The USCG is planning the sundown of the C-27J fleet and seeks to assess whether Dowty R391 propellers that have accumulated service on the C-27J (R391/6-132-F/10) can be overhauled and employed on the HC-130J (R391/6-132-F/3). OEM correspondence flags differences in usage spectra, airworthiness limitations, and platform designation/traceability. USCG seeks independent, rigorous industry input on whether a technical, airworthiness, and sustainment-acceptable path exists, with or without OEM endorsement.

Background and Technical Context

- Propeller variants:
 - HC-130J: Dowty R391/6-132-F/3 (P/N 697039001)
 - C-27J: Dowty R391/6-132-F/10 (P/N 697091001)
- Noted differences to be addressed:
 - Airworthiness limitations and inspection intervals differ across platforms (e.g., sampling and overhaul periods).
 - Usage spectra, mission profiles, maneuver spectra, and ground operation spectra differ between two- and four-engine platforms.
 - Type certificate listings and part numbering/traceability designate components to specific platforms.
 - System interfaces and TCTOs may affect loads and operations.
 - Engine Mount System Compatibility
- USCG intent: Evaluate feasibility of overhauled -F/10 shipsets or assemblies being installed on HC-130J, with no aircraft modification preferred, and with acceptable safety, fatigue life accounting, maintainability, and supportability.

Objective and Decision Questions

- Objective: Determine whether a technically acceptable, airworthy, and supportable path exists to use R391/6-132-F/10 units (post-overhaul) on HC-130J in lieu of R391/6-132-F/3.

- Decision questions:
 - Can structural and functional equivalence (or bounded non-equivalence with quantified margins) be demonstrated for HC-130J use?
 - Can validated life accounting and inspection intervals be established for cross-platform usage, including conversion of prior C-27J damage?
 - Can OEM or an acceptable airworthiness authority basis be maintained/obtained for continued airworthiness and life limits?

Scope of Work to Address in Responses

- Configuration and compatibility baseline:
 - Map -F/3 vs. -F/10 configurations: hub, blades, pitch change mechanism, valve housing, spinner, de-ice system, brush blocks/MPUs, harnesses, hardware.
 - Identify platform-specific part numbers, type certificate constraints, and traceability requirements; propose handling for cross-platform units.
 - Confirm physical interfaces: GMAD/gearbox attach, shaft length/offset, oil/electrical/de-ice connections, FADEC/EPCS expectations, prop brake interaction.
- Engine Mount System Compatibility Analysis
 - Respondents shall provide a comprehensive analysis of the engine mount system differences between the C-27J and HC-130J platforms and their impact on propeller dynamics and operational suitability.
 - Analyze differences in mount stiffness, damping characteristics, natural frequencies, and load transfer paths
 - Assess how mount system differences affect the coupled dynamics of the engine-gearbox-propeller assembly
 - Model the effect of different engine mount characteristics on propeller hub loads during normal operations, transients, and asymmetric thrust conditions
 - Quantify changes in vibratory load transmission from the propeller through the mount system to the airframe structure

- Evaluate potential for exciting new resonant frequencies or altering existing mode shapes in the propeller-engine-mount system
- Compare engine mount load paths and their influence on propeller attachment loads during various flight conditions
- Assess the impact of mount system differences on propeller shaft loads, particularly during engine start/shutdown transients and windmilling conditions
- Analyze how differences in mount compliance affect propeller control system loads and response characteristics
- Evaluate whether mount system differences necessitate different propeller operating limitations or procedures
- Assess the impact on propeller fatigue life accounting due to altered dynamic environment
- Identify any required modifications to propeller control system parameters to accommodate mount system differences
- Functional and performance assessment:
 - Pitch control range, governing stability, transient response, overspeed/underspeed protections.
 - Ground operations on four-engine aircraft: beta/reverse thrust usage, asymmetric taxi/turning, wind gust brake considerations.
 - Icing/de-ice performance under HC-130J conditions.
- Structural loads and fatigue life equivalency:
 - Develop mission-weighted usage spectra for USCG C-27J and HC-130J operations; include ground spectra (beta/reverse, crosswind taxi) and windmilling scenarios.
 - Component-level stress/life analysis for life-limited parts; cumulative damage methods (e.g., Miner's rule) with validated S-N/E-N and crack growth data.
 - Effects of four-engine aero/acoustic coupling, engine-out feathering vs. windmilling, adjacent prop interactions, and wing/nacelle dynamics.

- Propose severity/translation factors converting prior C-27J life consumption to HC-130J equivalent; include bounding cases and confidence treatment.
- Identify test instrumentation needs (strain/vibration on hub/blades/mounts; torque/power transients; nacelle/wing accelerometers) to validate analyses.
- System safety and airworthiness substantiation:
 - Perform hazard analyses (FHA, FMEA, FTA) for propeller-related failure conditions affected by model interchange.
 - Map proposed compliance to applicable criteria (e.g., MIL-HDBK-516C relevant sections, MIL-STD-1530C ASIP considerations, MIL-STD-882 system safety).
 - Define regulatory/acceptance path: if OEM endorsement is required/realistic; alternate government acceptance path and implications for sustainment/liability.
- Inspection, overhaul, and sustainment:
 - Overhaul/NDI/NDT tasks to bound unknown prior life and detect critical defects; supplemental inspections at installation and intervals thereafter.
 - Life limits and tracking approach within ACMS for cross-platform units; severity factors integrated; sampling/overhaul interval recommendations for HC-130J.
 - Supply chain and supportability effects: OEM support status, warranty, spares pooling, repair TAT, re-identification/recertification for platform designation.

In addition to the primary analyses requested, respondents shall provide a detailed assessment of the second-order technical effects resulting from the integration of the - F/10 propeller. This analysis must quantify the impact on systems and components not directly part of the propeller assembly but which are critically influenced by its operational characteristics.

- Propulsion System Dynamic Interaction Analysis:
 - Analyze the effects of any change in propeller mass, inertia, and aerodynamic damping on the entire powerplant rotating system. This must include a coupled torsional and dynamic analysis of the propeller, reduction gearbox (RGB), and engine driveshaft assembly.

- Assess the risk of exciting new, potentially damaging resonant frequencies or torsional modes within the gearbox and its components (e.g., gears, bearings, shafts).
- Model the impact on the Gearbox and Mounted Accessory Drive (GMAD) health and fatigue life, explicitly addressing how the new vibratory signature from the -F/10 propeller alters the certified loads spectrum for the GMAD.
- Define the engagement plan and data requirements needed from the powerplant OEM (Rolls-Royce) to validate this analysis.
- Aircraft-Level Vibration and Acoustic Signature Characterization:
 - Assess the change in vibratory loads transmitted from the nacelle into the wing structure. Quantify the impact on the fatigue life of primary and secondary wing structures, including the engine mount attachments and nacelle longerons.
 - Characterize the change in the far-field acoustic signature and the near-field structural-borne noise within the fuselage.
 - Evaluate the impact of the new vibration and acoustic environment on the reliability and performance of sensitive avionics and equipment located in the fuselage, particularly those near the propeller plane.
 - Assess the Human Factors impact of any new acoustic tones or vibrations on aircrew fatigue and communication clarity during representative USCG missions.
- Aerodynamic Interaction with Airframe and Control Surfaces:
 - Analyze the effect of altered propeller wash (slipstream) velocity, diameter, and swirl on the aerodynamic characteristics of the wing and empennage.
 - Assess any changes to wing lift distribution, stall characteristics, and minimum control speeds (V_{mcg} , V_{mca}), particularly during asymmetric thrust conditions (e.g., engine failure during takeoff).
 - Model the impact of the altered slipstream on the effectiveness and loading of control surfaces, including the rudder and elevator, during critical flight phases. This analysis should inform any required updates to the flight manual or aircrew procedures.
- Risk register and go/no-go criteria:

- Define objective acceptance thresholds for fatigue margins, dynamic stability, safety risk levels, inspection intervals, and supportability.

Government-Furnished Information (GFI) and Access

- The Government intends to provide or facilitate access to:
 - Applicable TCTOs and life-limit documents for -F/10 and -F/3 propellers and powerplants.
 - Maintenance manuals, overhaul work instructions, and service bulletin histories for both platforms.
 - Representative USCG mission usage data for C-27J and HC-130J operations.
 - Serialized histories for candidate C-27J propellers and subcomponents.
 - Points of contact for coordination with propeller/engine OEMs where possible.

Assumptions and Constraints

- No aircraft modifications are preferred; any proposed modifications must be explicitly identified, with cost/schedule/airworthiness impacts.
- Overhaul resets condition but not life-limited part counters; life accounting must be continuous, auditable, and compatible with ACMS.
- If OEM endorsement is a prerequisite in the proposed path, define engagement approach and expected artifacts. If not feasible, propose alternate acceptance path and associated risks.

Deliverables

- Technical plan and data gap assessment.
- Configuration and compatibility matrix (-F/3 vs. -F/10) with part number and traceability assessment.
- Usage spectrum characterization and translation methodology; severity factors with rationale.
- Structural life assessment report for life-limited parts; margins and recommended limits for HC-130J use post-overhaul.
- System safety assessment (FHA, FMEA, FTA) tailored to HC-130J context.

- Compliance matrix to applicable criteria (e.g., MIL-HDBK-516C, MIL-STD-1530C, MIL-STD-882).
- Overhaul/inspection recommendations for cross-platform units, including supplemental NDI/NDT.
- Test plan(s) and final reports if executed.
- Airworthiness path recommendation (OEM-supported vs. government acceptance); sustainment implications.
- Go/no-go recommendation with defined conditions and required mitigations.
- Data package for airworthiness review and ACMS updates.